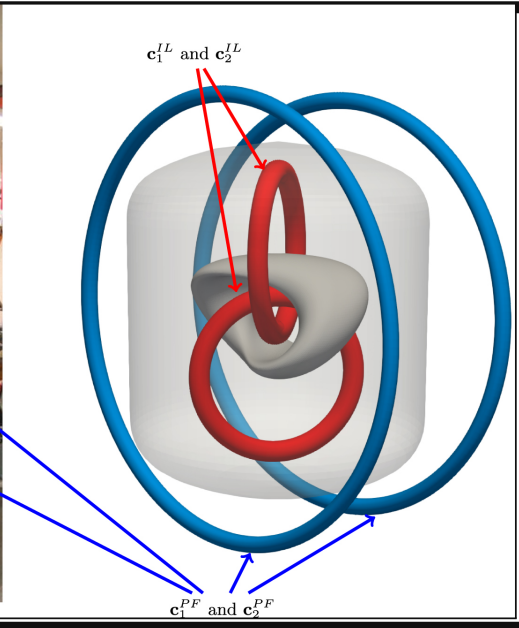


From Plasma Source Physics to Automated Shielding Design in Stellarators

Thaddaeus Kiker
MIT Plasma Science and Fusion Center
August 26th, 2026



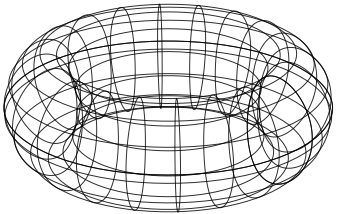

$\mathbf{c}_1^{IL}$ and $\mathbf{c}_2^{IL}$
$\mathbf{c}_1^{PF}$ and $\mathbf{c}_2^{PF}$

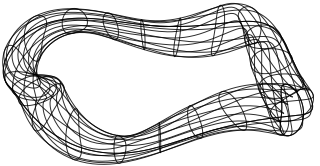
Contents

1. Native `StellaratorSource` in OpenMC
2. Spin-Polarized Plasma Neutronics
3. Automatic Reactor Shielding Optimization

Native `StellaratorSource` in OpenMC

Tokamaks vs. Stellarators: A Tale as Old as MCF





One (or many) poloidal sections to rule them ϕ all?

Building a General Stellarator Source in OpenMC

Peterson PR #3999

Native parametric `TokamakSource`

↓

`StellaratorSource`

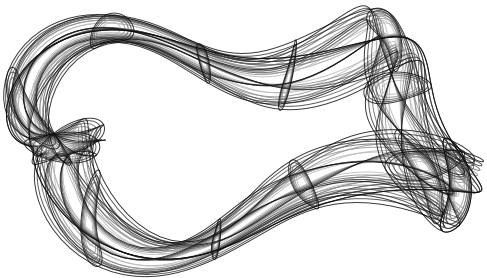
VMEC geometry and local field

The OpenMC source object, XML path, and Python API remain native.

The analytic Miller position model is replaced by an equilibrium-supplied three-dimensional grid.

Each sampled birth carries the local $\widehat{\mathbf{B}}$ needed by the polarized angular kernel.

Integrating VMEC (I/II): Three-Dimensional Plasma Coordinates



VMEC originated as a variational method for three-dimensional ideal-MHD equilibria.

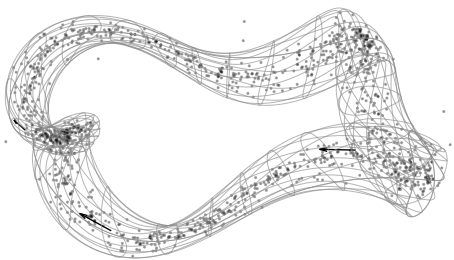
Its nested flux surfaces define

$$\mathbf{r}(\rho, \vartheta, \zeta)$$

and the physical volume element

$$\mathrm{d}V = \sqrt{g}\,\mathrm{d}\rho\,\mathrm{d}\vartheta\,\mathrm{d}\zeta.$$

Integrating VMEC (II/II): Neutron Birth and Local $\widehat{\mathbf{B}}$



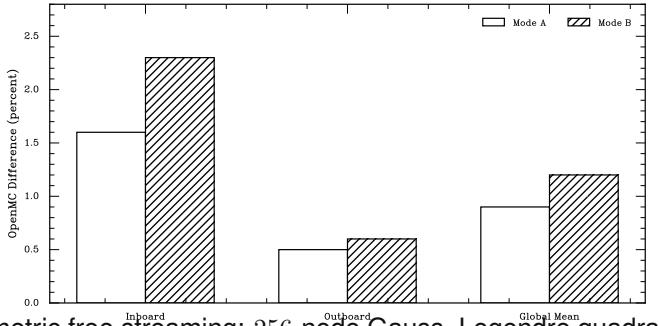
$$\mathrm{d}P \propto S(\rho)\,|\sqrt{g}|\,\mathrm{d}\rho\,\mathrm{d}\vartheta\,\mathrm{d}\zeta$$

$$p(\rho)\,p(\zeta \mid \rho)\,p(\vartheta \mid \rho, \zeta)$$

$$\widehat{\mathbf{B}} = \frac{\mathbf{B}}{|\mathbf{B}|}, \qquad w = 1$$

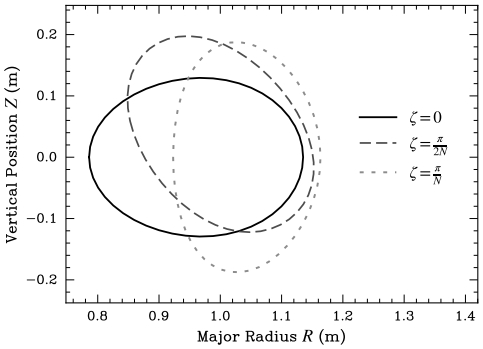
The conditional CDF cascade is rejection free.

StellaratorSource Validation (I/II)



- Axisymmetric free streaming: 256-node Gauss–Legendre quadrature agrees with Schwartz's `anarrima` closed forms.
- Real conformal wall: an independent point–patch sum gives 0.5–2.3% regional residuals and $0.914/0.920$ bin correlations.

StellaratorSource Validation (II/II)



QUASR QA 595509

II

Spin-Polarized Plasma Neutronics

Overview of Spin-Polarized Fusion (I/VI): Kulsrud's Claim

PHYSICS OF SPIN-POLARIZED PLASMAS

R.M. KULSRUD, E.J. VALEO, S.C. COWLEY
Plasma Physics Laboratory,
Princeton University,
Princeton, New Jersey,
United States of America

ABSTRACT. If the nuclear spins in a fusion reactor are oriented, or polarized, in an appropriate manner, then the nuclear reactions are modified in such a way as to enhance the performance of the reactor. The methods by which the spins can be polarized and the various difficulties connected with these methods are discussed. The processes by which the polarization can be lost owing to various physical processes in a confined plasma are assessed. The different polarization modes of the plasma ions are applicable to the different nuclear reactions, D–T, D–^{3}He and D–D. The benefits for reactor performance of these modes are discussed. It is concluded that on the basis of current knowledge, the possibility of achieving polarized spin in a fusion plasma cannot be ruled out and that the physics of such plasmas is well worth pursuing both for its intrinsic interest and for its benefit to fusion. Further, detailed calculations of the modification of the D–T nuclear reaction produced by spin polarization are given, as well as calculations of the depolarization rates for physical processes in a confined plasma. Finally, a kinetic description of a polarized plasma is outlined which generalizes the normal kinetic description of a plasma.

Overview of Spin-Polarized Fusion (II/VI): Nuclear-Spin States

n	$=$	$1, 2, \ldots$	Principal level (energy and shell size),
ℓ	$=$	$0, 1, \ldots, n-1$	Orbital L (orbital shell shape),
m_ℓ	$=$	$-\ell, \ldots, +\ell$	Orbital L_z (orbital spatial orientation),
m_s	$=$	$-\frac{1}{2}, +\frac{1}{2}$	Intrinsic-spin projection.

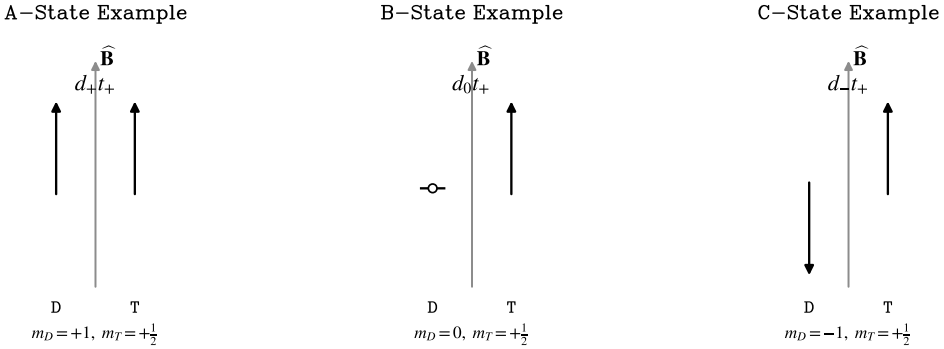
For a composite nucleus, angular momenta couple to total nuclear spin I and its field-axis projection m_I:

$$I_D = 1, \quad m_D \in \{-1, 0, +1\}; \qquad I_T = \frac{1}{2}, \quad m_T \in \left\{-\frac{1}{2}, +\frac{1}{2}\right\}.$$

With d_+, d_0, d_- and t_+, t_- denoting substate populations,

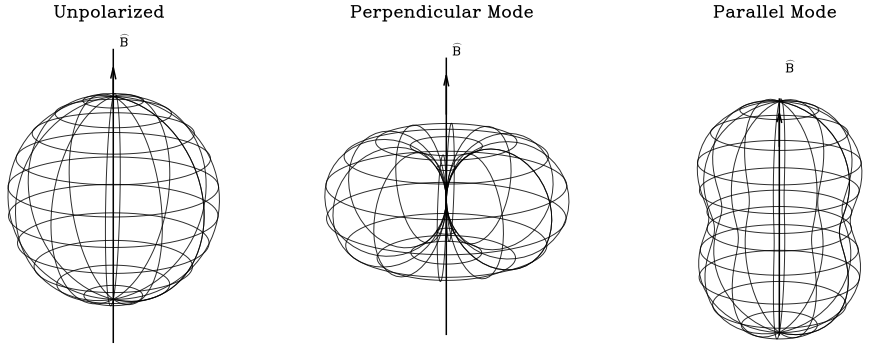
$$a = d_+ t_+ + d_- t_-, \qquad b = d_0 t_+ + d_0 t_-, \qquad c = d_+ t_- + d_- t_+, \qquad a + b + c = 1.$$

Overview of Spin-Polarized Fusion (III/VI): Spin States Along $\widehat{\mathbf{B}}$



The local field defines the quantization axis for prepared spin populations; it does not create high polarization by itself.

Overview of Spin-Polarized Fusion (IV/VI): Emission About $\widehat{\mathbf{B}}$



Polarizing the reacting fuel changes both the fusion yield and the neutron emission pattern relative to the local magnetic field.

Overview of Spin-Polarized Fusion (V/VI): Alpha Heating

A-State Reactivity: Same-sign D–T spin projections give

$$\frac{\sigma_A}{\sigma_0} = \frac{3}{2}, \qquad \frac{\mathrm{d}\sigma_A}{\mathrm{d}\Omega} \propto \sin^2\theta, \qquad \cos\theta = \widehat{\mathbf{B}} \cdot \widehat{\mathbf{u}}.$$

Alpha Self-Heating: More reactions produce more $3.5\ \mathrm{MeV}$ alpha particles. Their heating raises the plasma temperature, so burn feedback can increase fusion power by more than the 50% cross-section gain alone.

Overview of Spin-Polarized Fusion (VI/VI): Polarization Lifetime

A research program to measure the lifetime of spin polarized fuel

W. W. Heidbrink[1]*, L. R. Baylor[2], M. Büscher[3,4], R. W. Engels[3], A. V. Garcia[5], A. G. Ghiozzi[5], G. W. Miller[6], A. M. Sandorfi[7], X. Wei[8] and X. Zheng[7]

[1]Physics and Astronomy, University of California, Irvine, Irvine, CA, United States, [2]Oak Ridge National Laboratory, Oak Ridge, TN, United States, [3]Forschungszentrum Jülich, Jülich, Germany, [4]Department of Physics, Heinrich-Heine University Düsseldorf, Düsseldorf, Germany, [5]Oak Ridge Associated Universities, Oak Ridge, TN, United States, [6]Radiology and Medical Imaging and Biomedical Engineering, University of Virginia, Charlottesville, VA, United States, [7]Department of Physics, University of Virginia, Charlottesville, VA, United States, [8]Jefferson Laboratory, Newport News, VA, United States

The use of spin polarized fuel could increase the deuterium–tritium (D–T) fusion cross section by a factor of 1.5 and, owing to alpha heating, increase the fusion power by an even larger factor. Issues associated with the use of polarized fuel in a reactor are identified. Theoretically, nuclei remain polarized in a hot fusion plasma. The similarity between the Lorentz force law and the Bloch equations suggests polarization can be preserved despite the rich electromagnetic spectrum present in a magnetic fusion device. The most important depolarization mechanisms can be tested in existing devices. The use of polarized deuterium and ^{3}He in an experiment avoids the complexities of handling tritium, while encompassing the same nuclear reaction spin-physics, making it a useful proxy to study issues associated with full D–T implementation. ^{3}He fuel with 65% polarization can be prepared by permeating optically-pumped ^{3}He into a shell pellet. Dynamically polarized ^{7}Li-D pellets can achieve 70% vector polarization for the deuterium. Cryogenically-frozen pellets can be injected into fusion facilities by special injectors that minimize depolarizing field gradients. Alternatively, polarized nuclei could be injected as a neutral beam. Once injected, the lifetime of the polarized fuel is monitored through measurements of escaping charged fusion products. Multiple experimental scenarios to measure the polarization lifetime in the DIII-D tokamak and other magnetic-confinement facilities are discussed, followed by outstanding issues that warrant further study.

Spin-Polarized Fusion Changes Rate and Emission Angle

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\sigma_0}{2\pi}\left[\frac{3}{4}a\sin^2\theta + \left(\frac{2}{3}b + \frac{1}{3}c\right)\left(\frac{1}{4} + \frac{3}{4}\cos^2\theta\right)\right], \qquad \cos\theta = \widehat{\mathbf{B}}\cdot\widehat{\mathbf{u}}$$

Mode	(a, b, c)	Rate / Unpolarized Rate	Emission Relative to $\widehat{\mathbf{B}}$
Unpolarized	$\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$	1	Isotropic
A	$(1, 0, 0)$	$\frac{3}{2}$	Perpendicular
B	$(0, 1, 0)$	1	Parallel
C	$(0, 0, 1)$	$\frac{1}{2}$	Parallel

Prior Spin-Polarized Fusion Neutronics

Neutronics analysis of spin-polarized fuel in spherical tokamaks*

Jin Whan Bae[1,], Katarzyna Borowiec[1], Vittorio Badalassi[1], Jason Parisi[2], Ahmed Diallo[2], Jon Menard[2], Andrei Khodak[2] and Thomas Brown[2]**

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Accepted for publication 23 July 2025
Published 1 August 2025

Abstract

The effect of spin-polarized deuterium–tritium fuel on the blanket, magnets, and surrounding structures is analyzed for a spherical tokamak fusion power plant. The focus of this paper is the neutron flux distribution in the fusion device and the corresponding engineering considerations, such as tritium breeding ratio (TBR) and magnet fluence. Significant benefits were found for anti-aligned polarization: a 2.7% increase in the total TBR (which could increase significantly with further optimization) and an approximately 68% increase in the magnet lifetime. If the inboard breeding blanket can be replaced with extra shielding and magnets, then it could significantly increase fusion power with no loss of magnet lifetime. This would be accomplished by increasing the toroidal field strength.

Keywords: spin-polarization, tritium, fusion, spherical tokamak, neutronics

Analytic neutron wall loading from spin-polarized fusion in axisymmetric geometries

Jacob A. Schwartz[1,2]

[1]Previous affiliation: Princeton Plasma Physics Laboratory, Princeton NJ, USA
[2]Present affiliation: Marathon Fusion, San Francisco, CA, USA

July 17, 2025

Abstract

Spin-polarized nuclei can be used to enhance the fusion reaction rate, preferentially direct neutrons and alpha particles, or do some combination of these. In this paper we present formulas for the neutron wall load (NWL) on the walls of an axisymmetric torus with a convex cross section from filamentary ring sources with arbitrary fuel polarizations and arbitrary local magnetic field directions. While the NWL does not include neutron scattering, which would require a more sophisticated treatment, these formulas can be evaluated very quickly. This can help a fusion machine designer build intuition about the effects of spin-polarization on neutron wall loading. The formulas are also fully differentiable, so they could be included in a fusion systems optimization code. As an example, we present sets of first wall shapes optimized to have a uniform NWL for each of the main polarization modes.

Proposed StellaratorSource Interface for OpenMC

```
src = openmc.StellaratorSource(
    fluxmap="qh_fluxmap",
    polarization=(a, b, c),
    field_model="fluxmap",
    energy=openmc.stats.Discrete(
        [14.06e6], [1.0]),
    strength=1.0)
```

Compact Runtime Format

`.meta + .bin`

Python and XML use the existing source API

Isotropic behavior remains the default

Equilibrium conversion stays outside transport

One Kernel Supports Tokamak and Stellarator Sources

Tokamak

```
openmc.TokamakSource(
   major_radius=...,
   minor_radius=...,
   polarization=(a,b,c),
   field_model="pitched")
```

Miller birth-position sampler extended with toroidal or q-profile-pitched $\widehat{\mathbf{B}}$ and SPF emission.

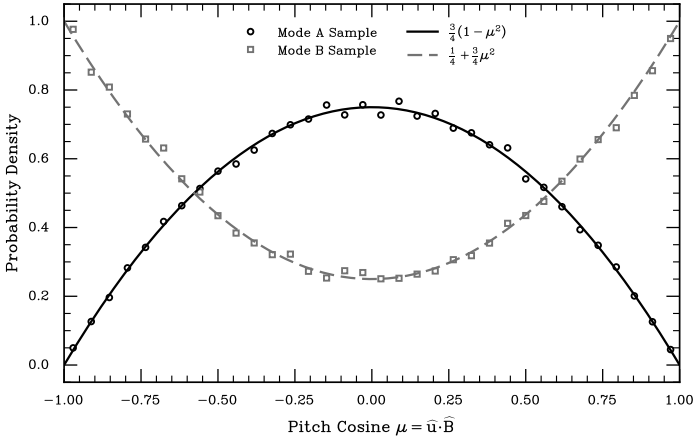
Stellarator

```
openmc.StellaratorSource(
   fluxmap="qh_fluxmap",
   polarization=(a,b,c),
   field_model="fluxmap")
```

VMEC position model with the gridded local magnetic field.

The angular mixture, inverse CDF, and three-dimensional rotation are shared.

SPF Kernel Validation (I/III)



SPF Kernel Validation (II/III)

$$G(\mathbf{w}) = \int_{V_p} \frac{S(\mathbf{r})\, f(\widehat{\mathbf{u}} \cdot \widehat{\mathbf{B}})}{4\pi \|\mathbf{w} - \mathbf{r}\|^2}\, \mathrm{d}V$$

OpenMC $\longleftrightarrow$ 256-node quadrature

Quadrature $\longleftrightarrow$ `anarrima`

A Mode	Inboard	Outboard
Schwartz text	$+43\%$	-22%
Quadrature	$+40.6\%$	-21.2%
OpenMC	$+39.2\%$	-21.0%

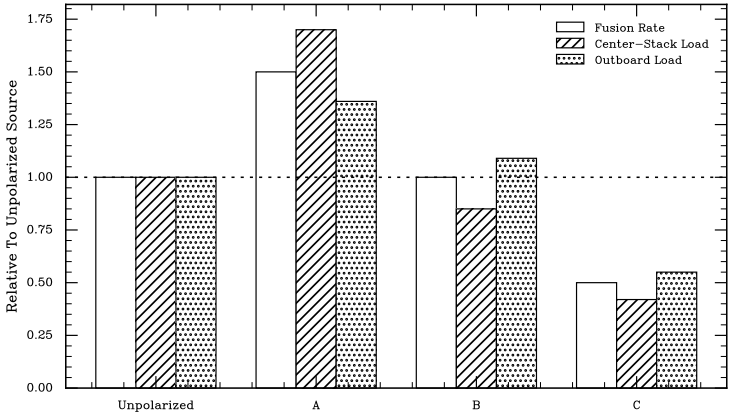
Convex, axisymmetric, free-streaming limit.

SPF Kernel Validation (III/III)

Schwartz (2025): in the fully polarized A state of an aspect-ratio-two box torus, the $\sin^2\theta$ kernel changes inboard and outboard NWL by $+43\%$ and -22%; quadrature and OpenMC recover both.

Bae et al. (2025): in an ARC-class FLiBe model, parallel emission raises TBR by 2.7% and lowers inboard magnet flux by 40%; the present model recovers both trends.

SPF Effects (I/II): R and NWL Coupling

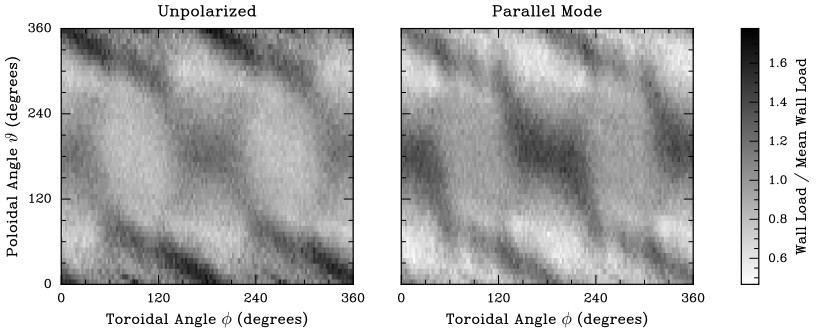


Mode A increases the reaction rate and compounds center-stack loading.

Mode B changes direction without a rate increase.

Mode C uses the B kernel at half the unpolarized rate.

SPF Effects (II/II): Poor Peaking Reduction by Geometry



Across fifteen usable QUASR equilibria, the peaking reduction has a 4.34% median, a 1.24%–5.67% interquartile range, and a 16.05% maximum.

III

Automatic Reactor Shielding Optimization

Nonuniform Shielding in Stellarator Design

ParaStell: parametric modeling and neutronics support for stellarator fusion power plants

Connor Moreno[1]*, Aaron Bader[1,2] and Paul Wilson[1]

[1]Computational Nuclear Engineering Research Group, Department of Nuclear Engineering and Engineering Physics, University of Wisconsin—Madison, Madison, WI, United States, [2]Type One Energy Group, Madison, WI, United States

The three-dimensional variation inherent to stellarator geometries and fusion sources motivates three-dimensional modeling to obtain accurate results from computational modeling in support of design and analysis of first wall, blanket, and shield (FWBS) systems. Manually constructing stellarator fusion power plant geometries in computer-aided design (CAD) and defining the corresponding fusion source can be cumbersome and challenging. The open-source parametric modeling toolset *ParaStell* has been developed to automate construction of such geometries in low-fidelity. Low-fidelity modeling is useful during the conceptual phase of engineering design as a means of rapidly exploring the design space of a given device. The modeling capability of *ParaStell* includes in-vessel components and magnets, for any given stellarator configuration, using a parametric definition and plasma equilibrium data. Furthermore, the toolset automates the generation of detailed, tetrahedral neutron source definitions and DAGMC geometries for use in neutronics modeling. *ParaStell* assists rapid design iteration, parametric study, and design optimization of stellarator fusion cores. As a demonstration of the design iteration capability, the effect of the three-dimensional parameter space on tritium breeding and magnet shielding is investigated, using the WISTELL-D configuration as a design basis. Blanket and shield thicknesses are varied in three dimensions, using the space available between the plasma edge and magnet coils as a constraint. The corresponding effects on tritium breeding ratio and magnet heating are tallied using the open-source Monte Carlo particle transport code OpenMC. The inclusion of additional and higher-fidelity modeling capabilities is planned for *ParaStell*'s future, as well as its implementation in machine-driven optimization.

DESIGNING ARIES-CS COMPACT RADIAL BUILD AND NUCLEAR SYSTEM: NEUTRONICS, SHIELDING, AND ACTIVATION

L. EL-GUEBALY,[1]* P. WILSON,[2] D. HENDERSON,[3] M. SAWAN,[2] G. SVIATOSLAVSKY,[2] T. TAUTGES,[3] R. SLAYBAUGH,[4] B. KIEDROWSKI,[4] A. IBRAHIM,[4] C. MARTIN,[4] R. RAFFRAY,[5] S. MALANG,[6] J. LYON,[7] L. P. KU,[7] X. WANG,[7] L. BROMBERG,[8] B. MERRILL,[8] L. WAGANER,[7] F. NAJMABADI,[1] and ARIES-CS TEAM

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[3]*University of California, San Diego, 9500 Gilman Drive, La Jolla, California 90093*
[4]*Consultant, Fliederweg 3, D 76351 Linkenheim-Hochstetten, Germany*
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[6]*Princeton Plasma Physics Laboratory, Princeton, New Jersey 08543*
[7]*Massachusetts Institute of Technology Plasma Science and Fusion Center Cambridge, Massachusetts 02139*
[8]*Idaho National Laboratory, Idaho Falls, Idaho*
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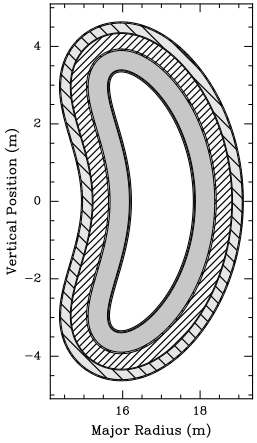
Received April 11, 2007
Accepted for Publication January 25, 2008

Within the ARIES-CS project, design activities have focused on developing the first compact device that enhances the attractiveness of the stellarator as a power plant. The objectives of this paper are to review the nuclear elements that received considerable attention during the design process and provide a perspective on their successful integration into the final design. Among these elements are the radial build definition, the well-optimized in-vessel components that satisfy the ARIES top-level requirements, the carefully selected nuclear and engineering parameters to produce an economic optimum, the modeling—for the first time ever—of the highly complex stellarator geometry for the three-dimensional nuclear assessment, and the overarching safety and environmental constraints to deliver an attractive, reliable, and truly compact stellarator power plant.

KEYWORDS: *stellarator, nuclear analysis, fusion power plants*

Note: Some figures in this paper are in color only in the electronic version.

Parameterizing Nonuniform Blanket and Shield Thickness



$$t_{\mathrm{br}} = t_{\mathrm{br}}(\vartheta, \phi), \qquad t_{\mathrm{sh}} = t_{\mathrm{sh}}(\vartheta, \phi)$$

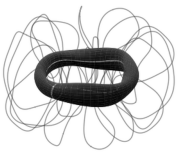
A design field changes the complete coil-response vector, TBR, shield mass, and maximum build.

The question is where material changes improve the limiting response without creating a new limit elsewhere.

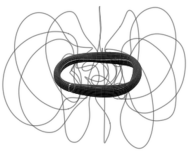
Searching for a Cheap Alternative to Neutronics (I/IV)



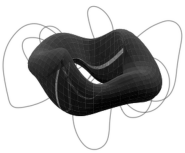
QUASR 1960314



QUASR 59509



QUASR 803097



QUASR 886079



QUASR 932746

Searching for a Cheap Alternative to Neutronics (II/IV)

Geometric Wall-Load Headroom

$$\mathrm{PF}_{\mathrm{geo}} = \frac{\max_{\mathbf{w}} G(\mathbf{w})}{\langle G \rangle_A}$$

$$G(\mathbf{w}) = \sum_v \frac{|\sqrt{g}_v|}{\|\mathbf{w} - \mathbf{r}_v\|^2}$$

Inverse-square geometry estimates wall-load headroom.

Effective Three-Dimensional Shaping

$$\varepsilon_{\mathrm{eff}} = \frac{\max_\phi \sqrt{\delta R^2 + \delta Z^2}}{\min_\phi |\Delta_0|}$$

Three-dimensional excursion is normalized by the minimum source–wall standoff.

Searching for a Cheap Alternative to Neutronics (III/IV)

Nematic Shape Coherence

$$Q = \left|\left\langle e^{2i\psi} \right\rangle\right|$$

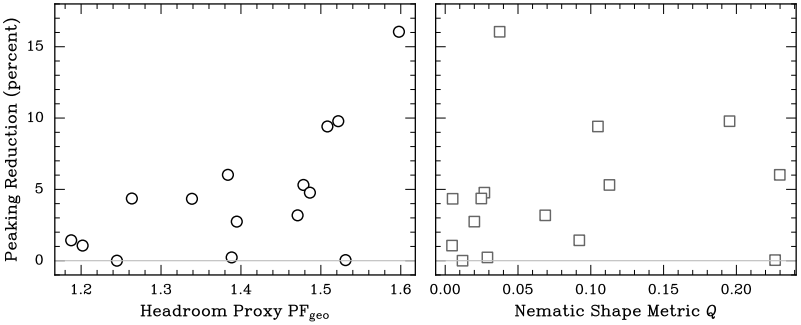
Wall-normal field angles share a preferred axis.

Winding-Pack Clearance

$$c(\vartheta, \phi) = \min_{\mathbf{x} \in \mathcal{C}} \|\mathbf{x} - \mathbf{r}(\vartheta, \phi)\|$$

Physical space available for added build.

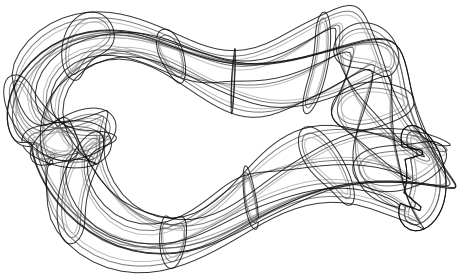
Searching for a Cheap Alternative to Neutronics (IV/IV)



Returning to Neutronics-Informed Automatic Optimization



Using ParaStell for Iterative Optimization



ParaStell regenerates the nonuniform breeder and shield surfaces for each candidate.

Initial contrast

$$\Delta t_{\mathrm{sh}} = -\Delta t_{\mathrm{br}}$$

at a fixed outer envelope.

Current comparison: composition swap, added build, and clearance-constrained redistribution.

Optimization Metric Selection (I/II)

Attribution	Worth	Clearance
$A(\mathbf{x}) = S(\mathbf{x})\psi^{\dagger}(\mathbf{x})$	$W_k = \int_{V_k} \phi\, \psi^{\dagger}\, \mathrm{d}V$	$c(\vartheta, \phi)$
Where harmful histories are born.	Where coil-important flux visits the existing build.	Where added material can physically fit.
Domain: plasma	Domain: shield	Domain: winding-pack geometry

Attribution cannot be back-projected directly into a shield-thickness update.

Optimization Metric Selection (II/II)

$$\underset{\mathbf{u},\eta}{\text{minimize}}\ \eta \qquad \text{subject to} \qquad R_j(\mathbf{u}) \le \eta \quad \text{for every coil } j$$

$$\mathrm{TBR}(\mathbf{u}) \ge \mathrm{TBR}_{\min}, \qquad M_{\mathrm{sh}}(\mathbf{u}) \le M_{\max}, \qquad t(\vartheta, \phi; \mathbf{u}) \le c(\vartheta, \phi)$$

Retain the complete response vector because reducing one coil can move the limiting response to another.

Compare equal-cost designs using peak magnet heating density, total cryogenic heating, and fast-neutron fluence.

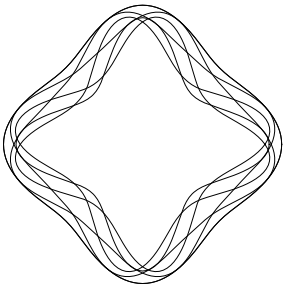
Choosing the Search Strategy

Gaussian-Process Bayesian Search	Empirical Response-Operator Update
8–20 smooth surface modes	Finite interventions estimate B_{FD}
Expected improvement under noisy, expensive OpenMC evaluations	A constrained minimax step selects the next candidate
Global exploration with explicit constraints	Full OpenMC corrects nonlinearity after each step

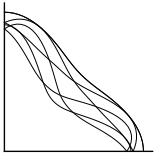
Both strategies keep the transport calculation in the loop.

Optimizing for Optimization

Full Torus



One Rotational Period



Rotationally periodic models keep histories that cross a field-period boundary (open wedges don't).

Working on Surface Source Read (SSR) implementation currently.

Early Optimization Caveats

$$\mathrm{FLiBe} \longrightarrow \mathrm{WC}$$

$$\Delta t_{\mathrm{sh}} = -\Delta t_{\mathrm{br}}$$

More high-density shield is added while neutron moderation and breeding volume are removed.

The fast component can increase even when total attenuation improves.

Multiple scattering connects one material change to several coils.

Plasma attribution therefore cannot be projected directly onto a shield thickness update.

Shielding Optimization: Current Status

End-to-End Design Loop Operational: ParaStell rebuilds nonuniform layers and OpenMC returns the complete coil-response vector with rotationally periodic transport.

Coarse Placement Sensitivity Resolved: A 12×20 intervention study identifies broad shield placements that reduce the limiting coil cluster—and others that increase loading.

Engineering Normalization Underway: Volume-normalized flux still peaks at coil 20; independent-seed reproduction, exact material mass, and clearance constraints are being finalized.

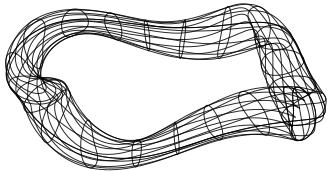
Next Steps for Optimization

Validate the Observed Control: Repeat the strongest interventions independently and evaluate magnet heating density, fast-neutron flux, TBR, and exact material cost.

Establish Spatial Convergence: Optimize successively finer, clearance-constrained design spaces until both performance and shield geometry stabilize.

Measure the Value of Response Awareness: Compare uniform, clearance-driven, wall-loading-driven, attribution-guided, and response-aware designs on equal-cost Pareto fronts.

Summary



A VMEC-native OpenMC stellarator source with independent source-space and transport-space validation.

One spin-polarized emission kernel shared by tokamak and stellarator sources.

A transport-in-loop shielding framework that now resolves placement sensitivity and has a defined path to an equal-cost constrained optimum.

IV

Backup Slides

Tokamak Local-Field Model

$$\widehat{\mathbf{B}} = \frac{\widehat{\boldsymbol{\phi}} + s\lambda\widehat{\mathbf{p}}}{\left\|\widehat{\boldsymbol{\phi}} + s\lambda\widehat{\mathbf{p}}\right\|}, \qquad \lambda = \frac{r}{q(\rho)R}, \qquad \rho = \frac{r}{a}.$$

r: Miller flux-surface minor-radius label at the sampled birth point.

a: edge minor radius; ρ is the normalized radius.

R: local cylindrical major radius, measured from the tokamak symmetry axis.

$q(\rho)$: safety factor, the toroidal turns of a field line per poloidal turn.

λ: local poloidal-to-toroidal field ratio, approximately B_p/B_ϕ.

$\widehat{\mathbf{p}}$: unit poloidal tangent; $\widehat{\boldsymbol{\phi}}$: toroidal unit vector; $s = \pm 1$: field handedness.

This approximation is used only by `TokamakSource`; `StellaratorSource` interpolates the cylindrical field components (B_R, B_ϕ, B_Z) from VMEC.

`StellaratorSource`: Computational Notes

`read_fluxmap()` $\longrightarrow$ `build_cdf_cascade()` $\longrightarrow$ `sample(seed)` $\longrightarrow$ `SourceSite`

Read-only state makes `sample()` thread-safe; position sampling is rejection free with unit weight.

Initialization

`spf_fluxmap_v1` loads a text `.meta` header and little-endian `.bin` arrays for (ρ, ϑ, ζ), $\sqrt{g}$, geometry, and cylindrical $\mathbf{B}$.

The constructor forms $S(\rho)|\sqrt{g}|$ and stores three normalized CDF tables: `cdf_rho_`, `cdf_zeta_`, and `cdf_theta_`.

Python writes XML; `polarization=None` omits SPF elements and retains the isotropic fallback.

Each Sample Call

`upper_bound_strided()` selects a cell in the order $\rho \to \zeta \to \vartheta$; three additional pseudorandom numbers jitter within it.

`interp()` performs trilinear interpolation: ρ is clamped, while ϑ and ζ wrap periodically.

Interpolated geometry gives the Cartesian birth site; interpolated (B_R, B_ϕ, B_Z) gives $\widehat{\mathbf{B}}$. Angle, energy, and time are then sampled.

SPF Emission Kernel: Symbol Reference

$$\eta = a + \frac{2}{3}b + \frac{1}{3}c, \qquad \frac{R_f}{R_{f,0}} = \frac{3}{2}\eta, \qquad P_\perp = \frac{a}{\eta}, \qquad x = \widehat{\mathbf{u}} \cdot \widehat{\mathbf{B}} = \cos\theta.$$

a: same-sign D–T spin projections; b: zero deuteron spin projection, $m_D = 0$; c: opposite-sign projections. The populations satisfy $a + b + c = 1$.

η: total-rate coefficient; $R_f/R_{f,0}$ is the fusion-rate multiplier relative to unpolarized fuel.

$P_\perp$: probability of selecting the perpendicular angular kernel.

$\widehat{\mathbf{u}}$: sampled neutron birth direction; $\widehat{\mathbf{B}}$: local magnetic-field direction; x: pitch cosine relative to the field.

$$p_\perp(x) = \frac{3}{4}(1 - x^2), \qquad p_\parallel(x) = \frac{1}{4} + \frac{3}{4}x^2.$$

A mode $(1, 0, 0)$ gives 1.5 times the unpolarized rate and perpendicular emission. B and C use the parallel kernel.

SPF Kernel: Computational Notes

$$\texttt{polarization} \longrightarrow p_\perp \longrightarrow \cos\theta \longrightarrow \widehat{\mathbf{u}}_{\mathrm{local}} \longrightarrow \widehat{\mathbf{u}}_{\mathrm{global}}$$

Per-neutron sampler: `SourceSite::wgt` is unchanged; absolute rate uses source strength or postprocessing.

Initialization

XML (a, b, c) is validated and normalized; the constructor stores $p_\perp = a/\eta$. If absent, the original isotropic path is used.

Field direction: `TokamakSource` uses `toroidal` or q-profile `pitched`; `StellaratorSource` uses `fluxmap` or a toroidal cross-check.

Both sources call the same `sample_polarized_direction()` routine.

Each Direction Draw

Fixed PRNG order: mode selector, inverse-CDF uniform variate, then local azimuth. This enables fixed-seed parity tests.

The perpendicular and parallel $\cos\theta$ laws use exact cubic inverse CDFs, eliminating angular rejection loops.

Least-aligned-axis Gram–Schmidt avoids parallel-axis degeneracy before global rotation.

Adjoint Computation

$$L^{\dagger}\psi_j^{\dagger} = q_j, \qquad A_j(\mathbf{r}) = \int S(\mathbf{r}, E, \Omega)\, \psi_j^{\dagger}(\mathbf{r}, E, \Omega)\, \mathrm{d}E\, \mathrm{d}\Omega, \qquad R_j = \int A_j(\mathbf{r})\, \mathrm{d}V.$$

$L^{\dagger}$: adjoint transport operator; it propagates response importance backward through the reactor.

q_j: coil-j response source: fast flux or copper kerma (energy transferred per mass).

$\psi_j^{\dagger}$: importance of a neutron state $(\mathbf{r}, E, \Omega)$: position, energy, and direction.

Scope: one adjoint gives attribution for one response, not a shield-material derivative.

S: plasma birth source; A_j: where histories responsible for response j are born; R_j: scalar coil response.

OpenMC method: multigroup random ray (deterministic track-based transport) with a filamentary adjoint source along the target coil.

Reciprocity: $\int S\psi_j^{\dagger}$ reproduced the forward coil ranking (Spearman rank 0.76 for QH; 0.83 for QA).

FW-CADIS and Weight Windows

Forward-Weighted Consistent Adjoint-Driven Importance Sampling

forward random-ray flux $\psi \longrightarrow$ adjoint importance $\psi^{\dagger} \longrightarrow$ weight windows (w_L, w_U).

w_L, w_U: particle-weight bounds. Histories above w_U split; those below w_L undergo Russian roulette. Correct accounting preserves the expected tally while moving effort to important phase space.

What Ran: A one-period DAGMC (CAD-based geometry) pilot produced a $10 \times 10 \times 10$ map. Forward and adjoint source-region miss rates were 19.38% and 43.39%: an integration result, not a production map.

What Was Used: The 12×20 matrix used analog forward transport without weight windows. The direct adjoint supplied attribution separately.

Surface Source Read (SSR)

Method

Write: bank each particle crossing a selected surface, including position $\mathbf{r}$, direction $\hat{\mathbf{u}}$, energy E, statistical weight w, and time t.

Read: relaunch that bank through a changed downstream geometry instead of resampling the plasma.

Exactness requires a clean separation between unchanged upstream transport and the varied downstream problem, without lost or double-counted crossings.

First-Wall Test

A first-wall bank kept breeder and shield downstream and therefore editable.

The nonconvex torus produced 2.36 records per birth; 58% were scattered recrossings. Fresh-history filtering recovered the inboard response to 4.1% CV (coefficient of variation).

Far coils remained 3.5–5 times low; replay saved only 11–13% because blanket and shield scattering still ran.

Status: not used for the current intervention matrix.

Alpha Self-Heating Beyond the Cross-Section Gain

$$D + T \longrightarrow \alpha\,(3.5\ \mathrm{MeV}) + n\,(14.1\ \mathrm{MeV}), \qquad P_\alpha = \frac{3.5}{17.6} P_f \approx 0.20 P_f$$

$$\sigma_{DT} \uparrow \Longrightarrow\ P_f \uparrow \Longrightarrow\ P_\alpha \uparrow \Longrightarrow\ T_i \uparrow \Longrightarrow\ \langle \sigma v \rangle \uparrow$$

The A-state $\sin^2\theta$ kernel biases alpha birth velocity perpendicular to $\widehat{\mathbf{B}}$, but more $v_\perp$ does not make the gyro-orbit tighter:

$$r_{L\alpha} = \frac{m_\alpha v_\perp}{2eB}.$$

Anisotropic-alpha confinement requires a fast-ion orbit calculation. The established greater-than-50% power gain is the nonlinear self-heating feedback above.

Preparing and Losing Nuclear-Spin Polarization

Prepare and Deliver

Deuterium: Dynamic nuclear polarization in cryogenic LiD transfers electron order to deuterons; demonstrated vector polarization reaches about 70%.

^{3}He: Spin-exchange optical pumping precedes shell permeation; reported polarization reaches about 65%.

Delivery: Cryogenic injection and a continuous guide field preserve the axis. A polarized neutral beam is the principal alternative.

Depolarize

Field Nulls: With no field, the reference axis is lost; injection therefore requires a guide field.

Bound Electrons and Surfaces: Hyperfine coupling during ionization, ablation, or recycling can scramble nuclear spin.

Resonant Fields: Waves near the nuclear precession frequency can drive spin flips. Core Coulomb collisions and nonresonant turbulence are predicted to be weak.

The DIII-D Program Will Measure Polarization Survival

$$D + {}^{3}\mathrm{He} \longrightarrow \alpha\,(3.6\ \mathrm{MeV}) + p\,(14.7\ \mathrm{MeV})$$

Prepare the Proxy Fuel: Polarized LiD and ^{3}He pellets reproduce the low-energy D–T spin physics without handling tritium.

Infer the Lifetime: The poloidal and pitch distributions of escaping protons and alphas retain information about the reactants' polarization state.

Program Status: By the 2025 APS-DPP update, Jefferson Lab had commissioned the LiD irradiation beamline while UVA, ORNL, and UCI were developing ^{3}He pellets, cryogenic injection, plasma scenarios, and charged-product diagnostics. No polarization-lifetime result had yet been reported.